SpringForge

Intelligent Spring Boot Development Ecosystem

Spring Forge

The

Developers waste 50-70% of time on manual architecture setup, boilerplate coding, quality checks, and CI/CD configuration instead of building features.

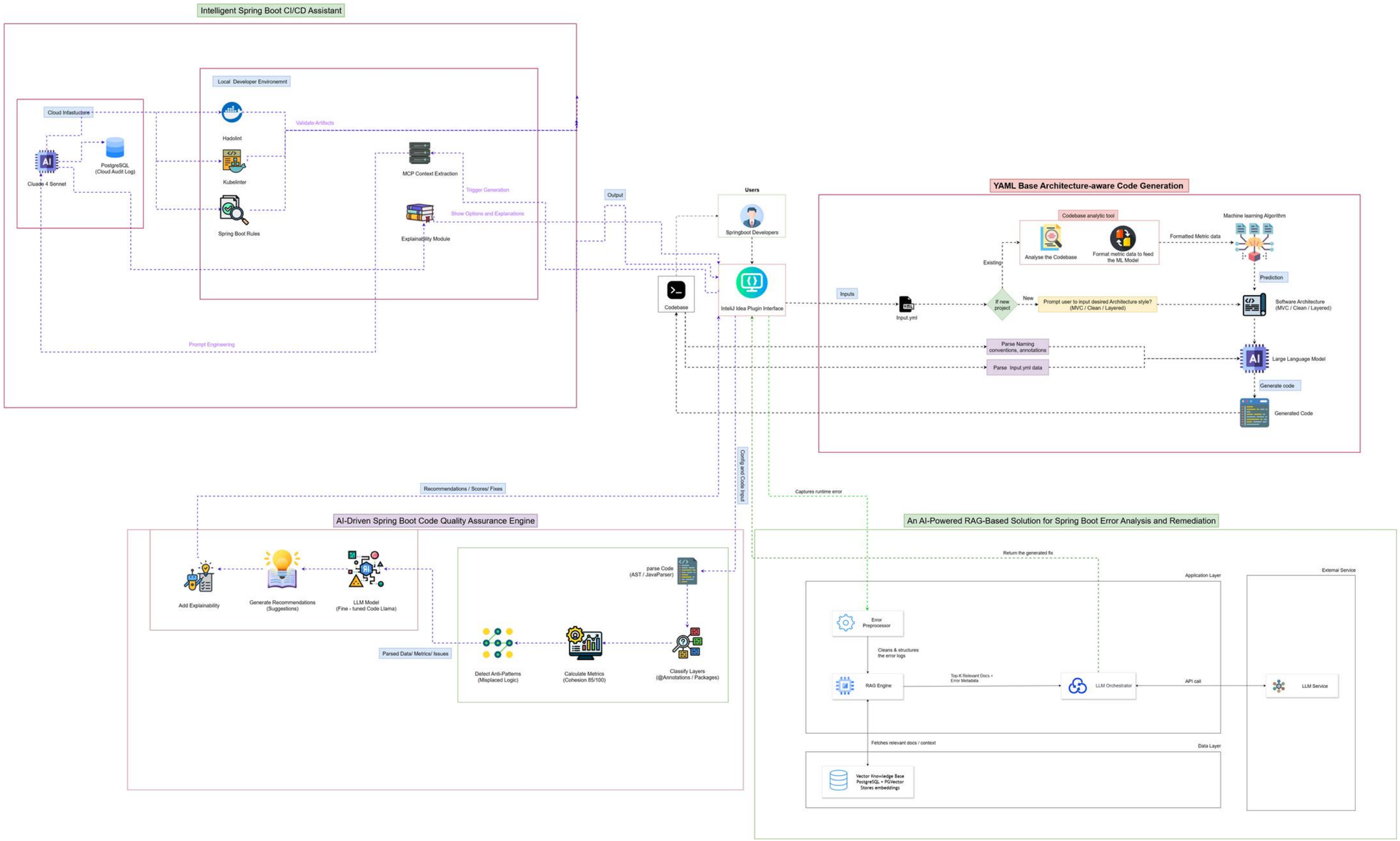
Impact/

- Time Waste
- Quality Issues
- Deployment Delays
- Cost

Benefits

- Springboot developers

Overall System Diagram



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ARCHITECTURE-AWARE Spring Boot Code Generation Plugin

Architecture-Aware Spring Boot Code Generation Plugin

The Sub

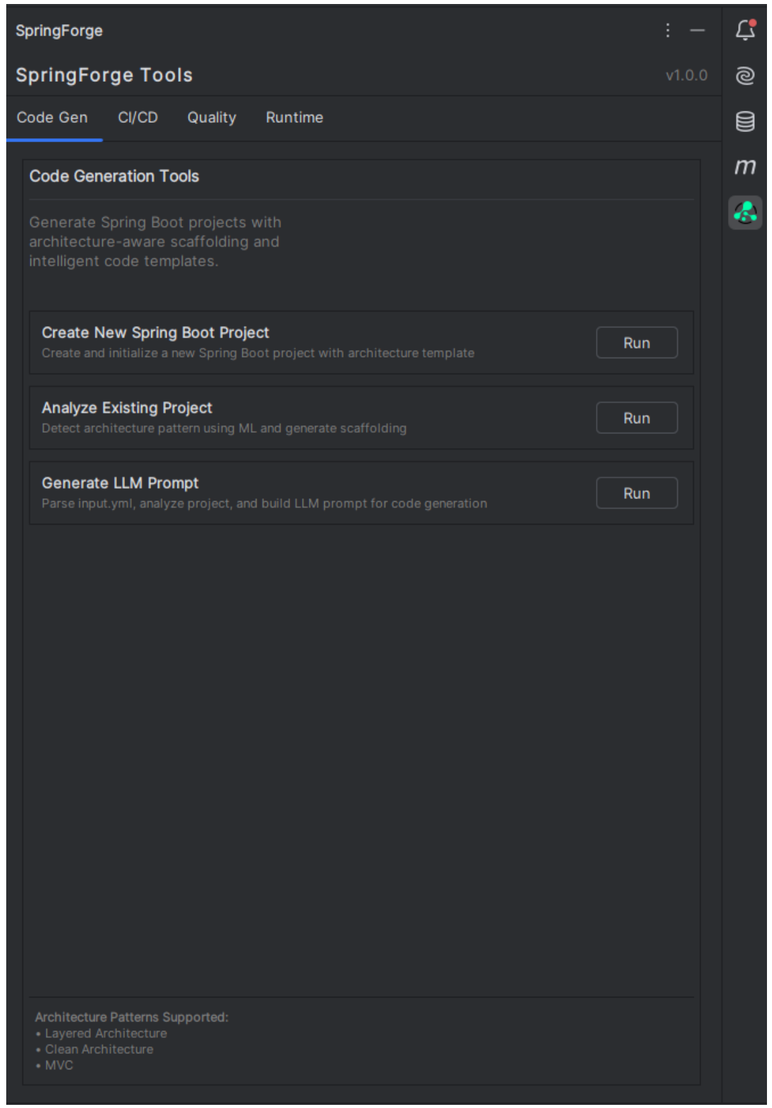
Not Having a proper scaffolding tool for Spring Boot and Generic code generators ignore project-specific architecture leading to inconsistent code.

How solution will

- Addresses the problem of manual, inconsistent Spring Boot project scaffolding
- Automatically analyzes existing project structure and detects architecture (Layered, MVC, Clean) using a trained machine learning model.
- The plugin analyzes the active project, extracts specific naming conventions and folder structures, and combines them with user input and predicted architecture to generate proper codes.
- Reduces setup time and architectural mistakes during early development

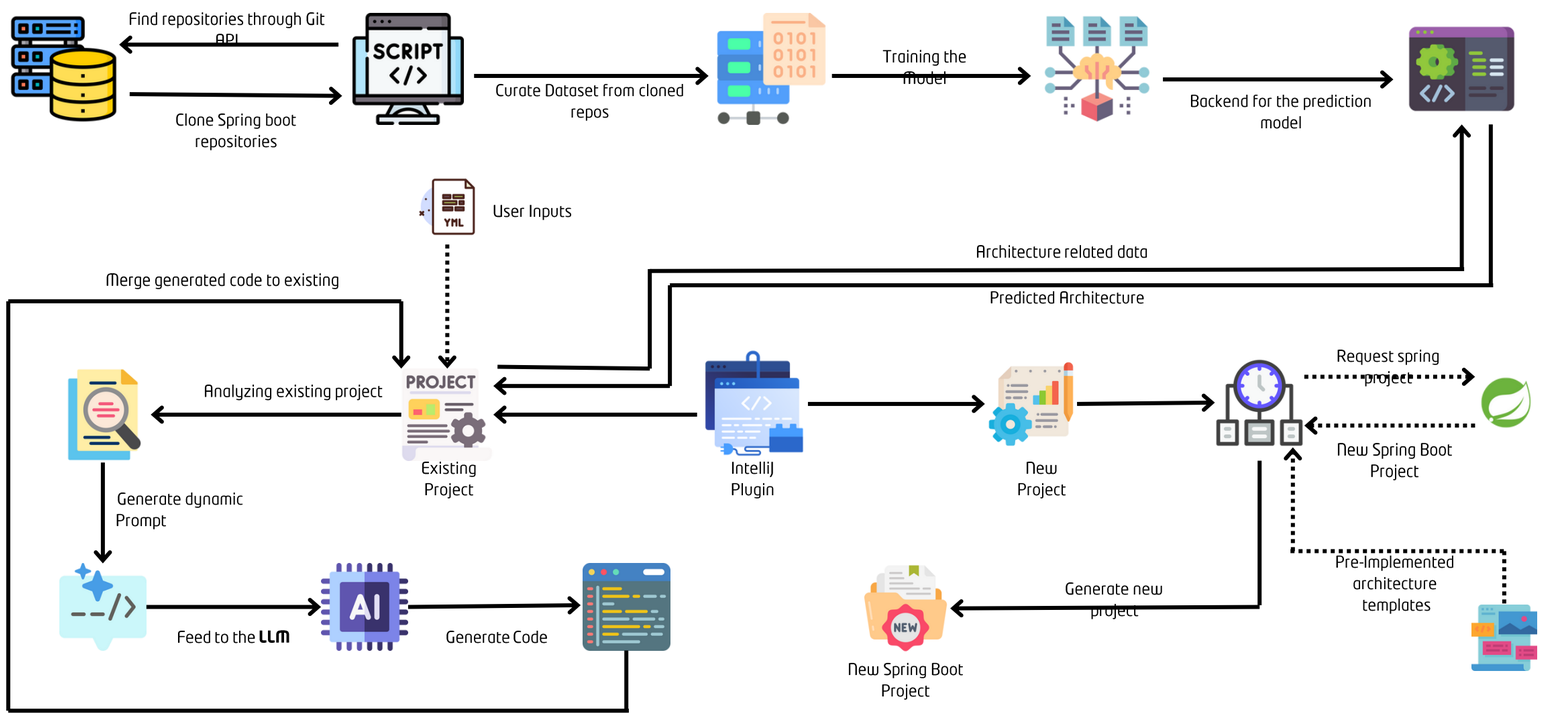
User Requirements

- Eliminates manual boilerplate setup by integrating a custom Spring Initializr wrapper directly in the IDE to create projects with pre-defined architectural structures (Layered/Clean or MVC)
- Maintain consistent architecture and naming conventions
- Workflow Integration: Fulfills the requirement for an IDE-native experience; users analyze projects and generate code without leaving IntelliJ.



Architecture-Aware Spring Boot Code Generation Plugin

Design Excellence



- IntelliJ Plugin–based solution integrated directly into developer workflow.
- Hybrid Analysis Engine: A novel contribution combining Static Analysis to extract code features with Machine Learning to classify the project architecture before generating code.
- Dynamic Prompt Construction: Unlike standard tools that use static templates, SpringForge dynamically builds prompts injecting live project constraints (package depth, annotation usage, software architecture) to ensure LLM output is structurally valid.
- Decoupled Microservice design where the heavy ML processing is offloaded to a Cloud/FastAPI service, keeping the IntelliJ Plugin lightweight and responsive.

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An AI-Powered RAG-Based Solution for Spring Boot Runtime Error Analysis and Remediation

An AI-Powered RAG-Based Solution for Spring Boot Runtime Error Analysis and Remediation

Problem

Developers waste 30-40% of debugging time context-switching between IDE, Stack Overflow, and documentation. Generic search returns irrelevant results.

User

IDE-integrated RAG system that provides AI-powered debugging assistance without leaving the development environment.

Functional

- Plugin should be able to provide tool window for runtime error input.
- Plugin should be able to automatically extract relevant code files.
- Plugin should support cosine similarity search on a vector database to retrieve semantically relevant entries.
- Plugin should display AI-generated responses clearly within the tool window.

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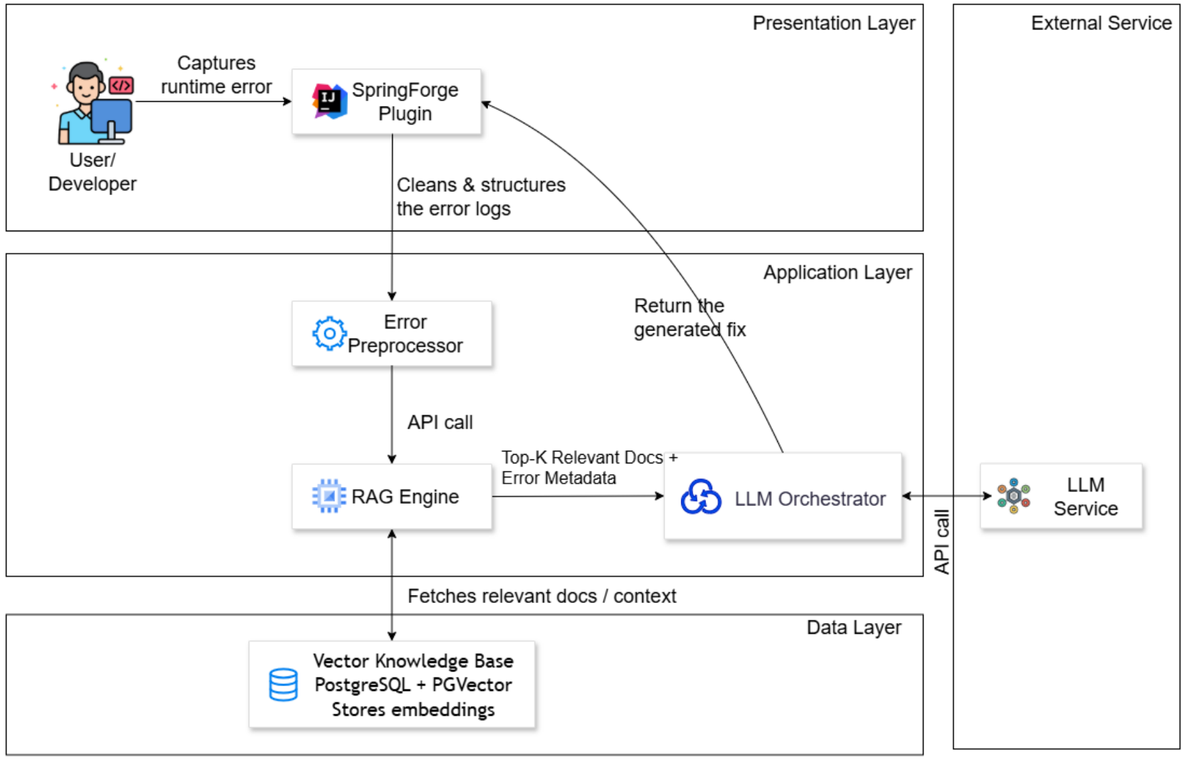
- Plugin should be able to process requests and return results within 10 seconds.
- Plugin should be able to achieve >75% precision in retrieving relevant documentation
- Plugin should be able to generate correct fixes with >90% accuracy

An AI-Powered RAG-Based Solution for Spring Boot Runtime Error Analysis and Remediation

Design Excellence



Novel Contribution: AI powered RAG-based Spring Boot debugger with IDE integration

Design Aspect	Innovation	Impact/Result
Semantic understanding	Vector embeddings	80% precision
Grounded AI	RAG module	Reduces hallucination
Performance optimization	Lightweight model + async pipeline	8.8s response

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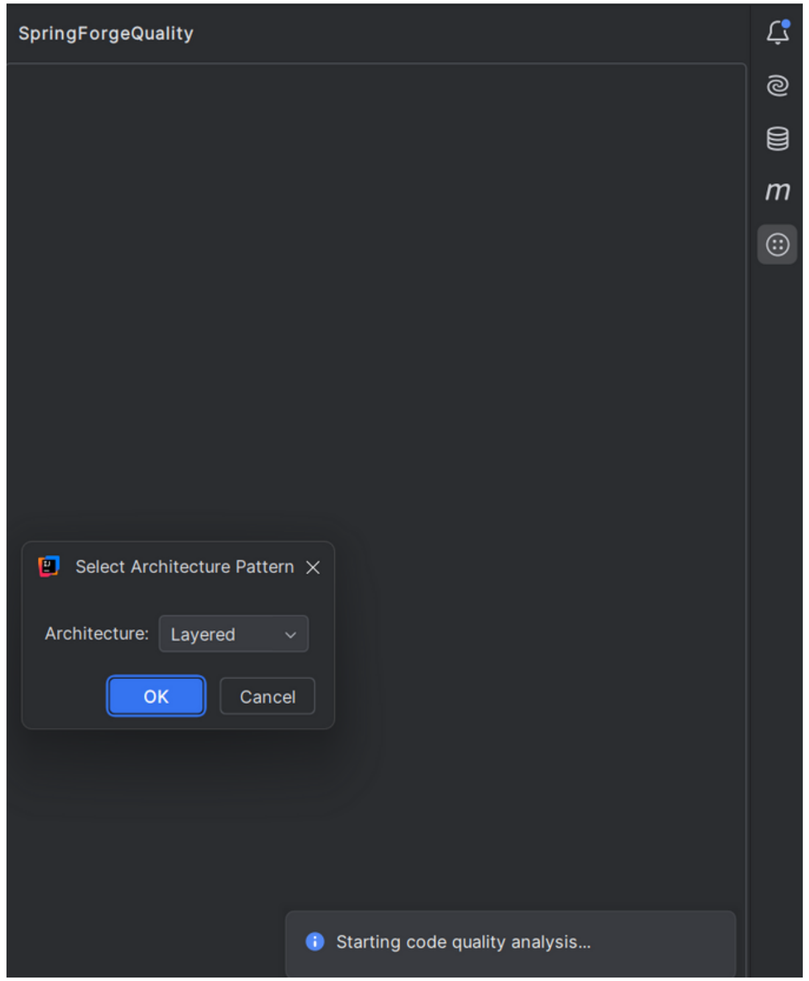
AI-Driven Code Quality Spring Boot Assurance Engine As A Plugin

AI-Driven Code Quality Spring Boot Assurance Engine As A Plugin

The Sub Problem

Spring Boot developers lack tools to automatically detect architecture violations and measure code quality. Manual reviews miss hidden anti-patterns, causing inconsistent structure, poor maintainability, and increased debugging effort within large or evolving projects.



How solution will

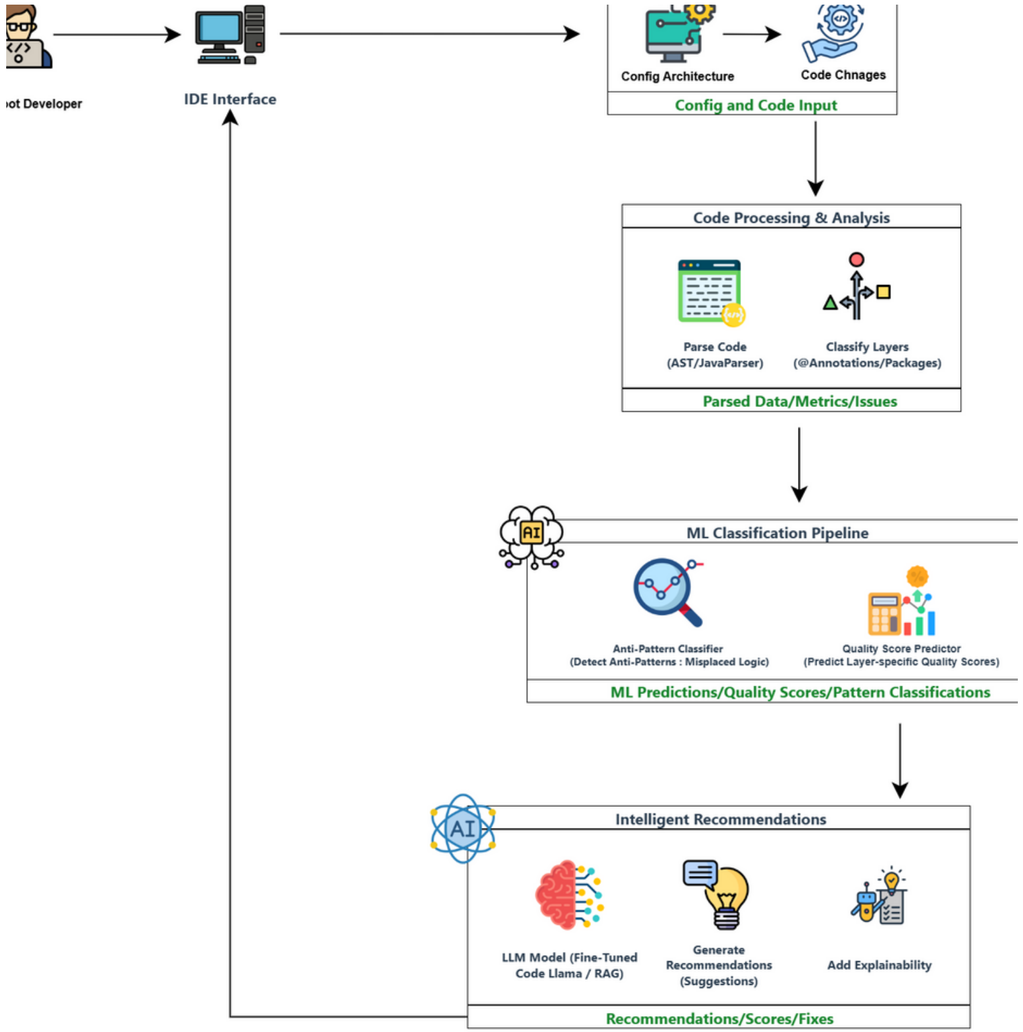
- IntelliJ plugin automatically analyzes the current Spring Boot project.
- Uses a trained ML model to detect architecture issues.
- Uses a second ML model to generate a 0–100 code quality score based on architecture rules.
- Automatically extracts project features (lines of code, imports, annotations, dependencies, complexity, layers) using IntelliJ PSI.
- Shows instant feedback inside IntelliJ—no external tools needed.
- Provides AI-generated explanations and recommended fixes to improve architecture and code quality.

User Requirements

- Generates a clear 0–100 code quality score, helping teams measure structural health and maintain long-term code consistency.
- Produces simple explanations and recommended corrections, enabling faster debugging and more maintainable code.
- Provides all analysis, warnings, and reports inside IntelliJ's tool window—no need to switch applications or use external scanners.

AI-Driven Code Quality Spring Boot Assurance Engine As A Plugin

Design Excellence



1. Dual-Engine AI Architecture

- Anti-Pattern Classifier + Quality Score Regression
- Detects violations and quantifies code quality (0–100)

2. IntelliJ-Native Smart Analysis

- Uses PSI-based static analysis
- Real-time extraction of LOC, dependencies, annotations, complexity
- Results shown instantly inside a custom tool window

3. Architecture-Aware Intelligence

- Understands Layered/Hexagonal/Clean/MVC constraints
- Detects context-specific issues (business logic placement, missing transactions, tight coupling, etc.)

4. Actionable Developer Feedback

- Explains why the issue happened and how to fix it
- Makes the ML output understandable and immediately usable

5. Deployment-Ready ML Service

- Clean FastAPI microservice with schema validation
- Model, encoder, and feature mapping fully encapsulated

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Intelligent Spring Boot CI/CD Assistant

AI-Powered Infrastructure Generation with Security Validation and Explainability

The Sub

Lack of secure, Spring Boot-aware CI/CD artifact generation with existing tools producing insecure infrastructure code lacking validation and explainability.

How solution will

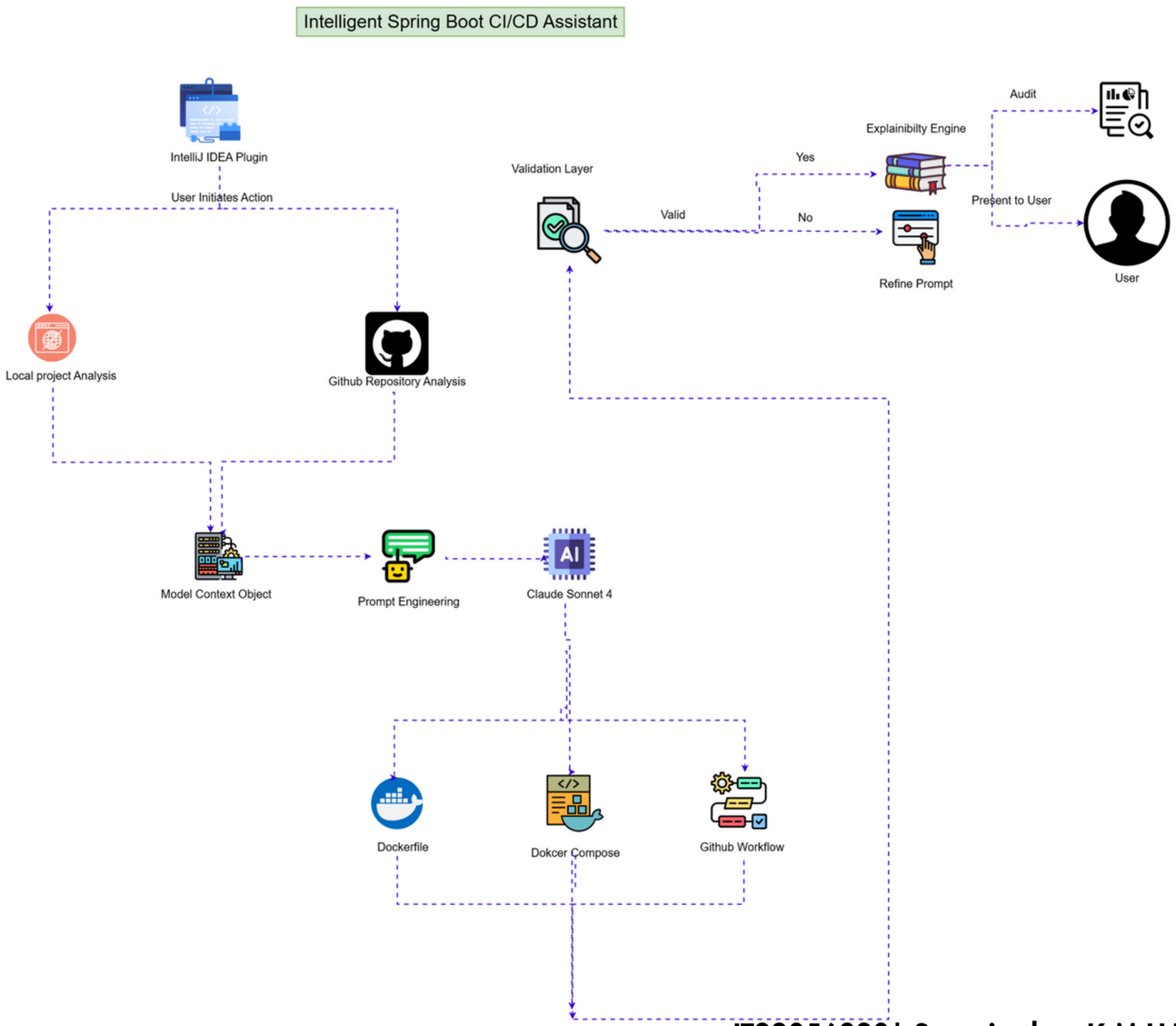
- Addresses the problem of manual, error-prone, and insecure DevOps artifact creation for Spring Boot projects
- Automatically analyzes project context (dependencies, build tools, configurations) using Model Context Protocol (MCP)
- Generates secure, production-ready deployment artifacts (Dockerfiles, Docker Compose, GitHub Actions) with integrated multi-layer validation (Hadolint, KubeLinter, Spring Boot rules) and line-by-line explainability for DevSecOps education
- Reduces deployment configuration time by ≥50% and eliminates common security vulnerabilities

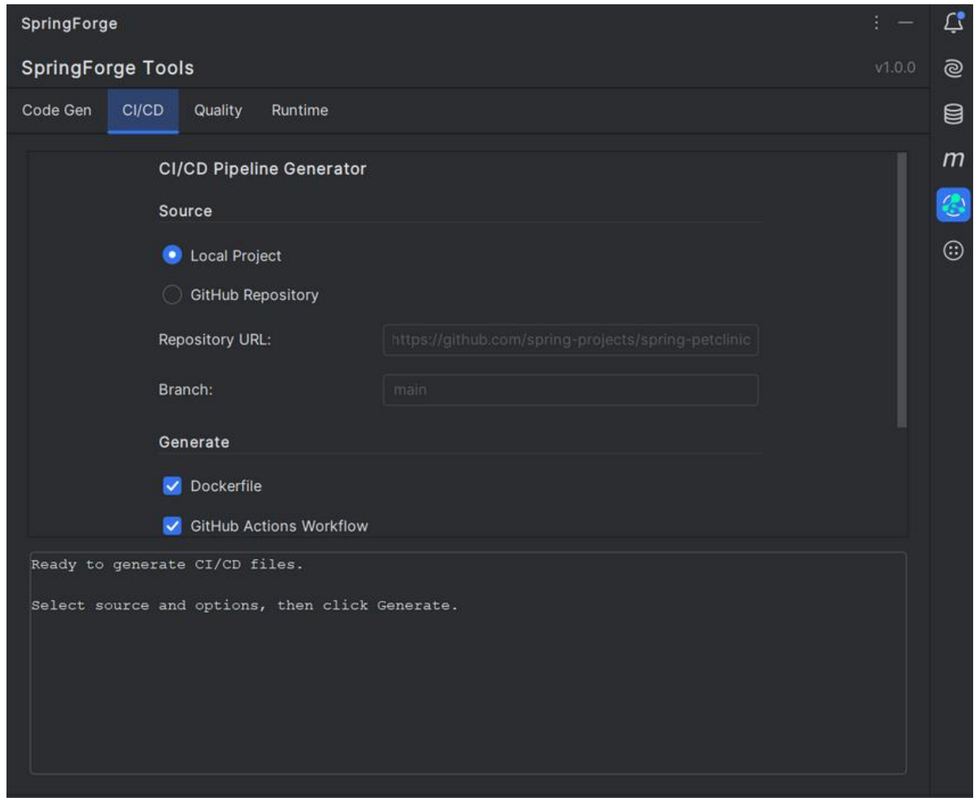
User Requirements

- Ability to generate deployment artifacts without manual DevOps expertise
- Automated security validation preventing hardcoded secrets, root-level containers, and missing resource limits
- Context-aware generation matching specific project requirements (Java version, dependencies, ports)
- Educational explanations for junior developers learning containerization and CI/CD
- Seamless integration inside IntelliJ IDEA with audit logging for compliance and traceability

AI-Powered Infrastructure Generation with Security Validation and Explainability

Design





Commercialization Strategy

Customer

- Spring Boot developers struggling with architecture decisions, code quality, and DevOps setup
- Junior Developers & Newcomers

Market

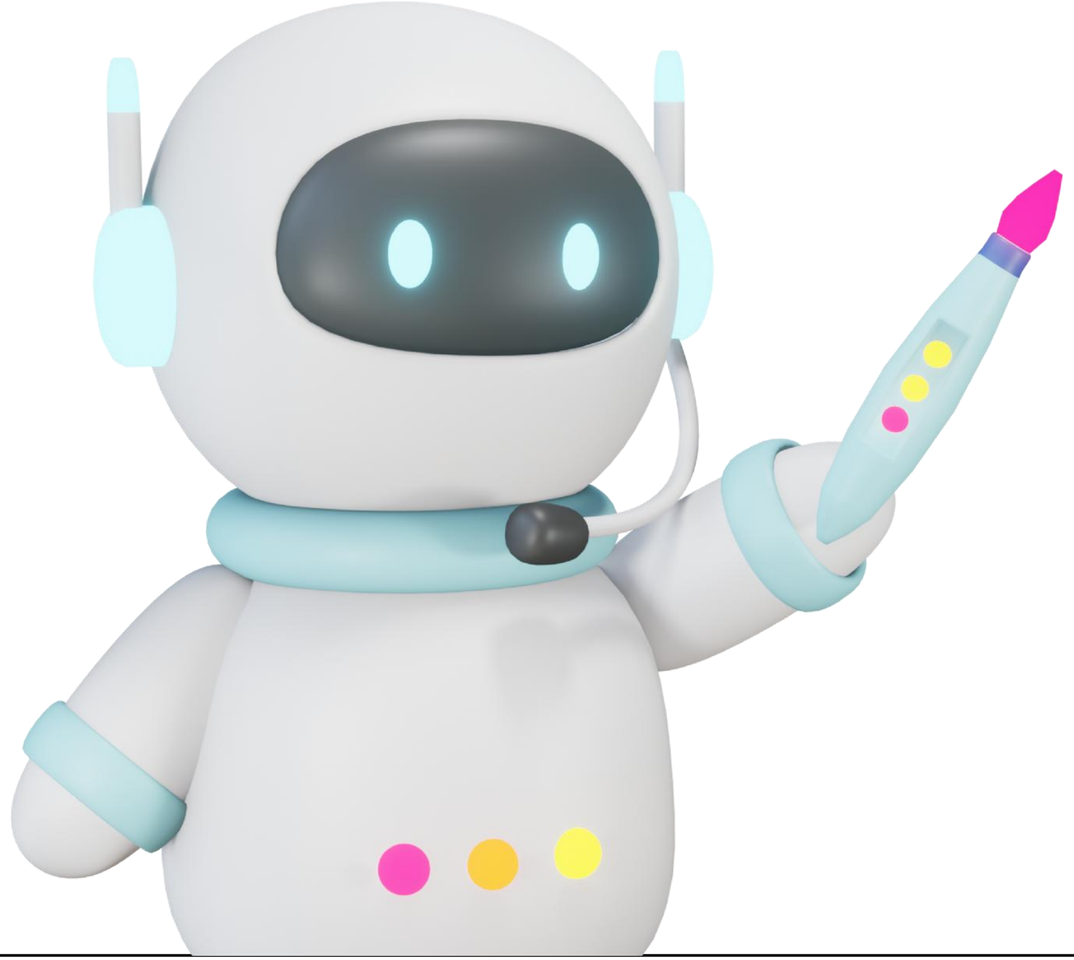
- Out of 8.7M Spring Boot developers, we're targeting around 10%.

Unique

- AI-Powered Architecture Detection
- Dual-Source Analysis
- End-to-End Automation
- Context-Aware Intelligence
- Zero-Vendor Lock-in

Revenue & Cost

- Revenue Model: Freemium Subscription.
- Community: Free
- Professional and Enterprise editions with paid subscription with enhanced features



Thank You